

Abel's Theorem

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Abstract. Write your abstract here. State the problem, the contribution, and the main result in 4–8 sentences.

Keywords: keyword1, keyword2

1 Abel's Lemma

Lemma 1.1 (Abel's Lemma). *Assume that the power series*

$$\sum_{k=0}^{\infty} a_k x^k$$

converges for some $x = x_0 \neq 0$. Then the series converges absolutely for $x \in \mathbb{R}$ such that $|x| < |x_0|$

Proof.

$$\sum_{k=0}^{\infty} a_k x^k \text{ converges} \Rightarrow \lim_{k \rightarrow \infty} a_k x^k = 0$$

and in particular $|a_k x_0^k| \leq C$ for $k = 1, 2, \dots$. Then

$$|a_k x^k| = |a_k x_0^k| \left| \frac{x}{x_0} \right|^k \leq C \left| \frac{x}{x_0} \right|^k.$$

Note $\left| \frac{x}{x_0} \right| < 1$ thus it is a geometric series and hence converges.

$$\sum_{k=0}^{\infty} a_k x^k \text{ converges absolutely}$$

□

2 Main Theorem

Theorem 2.1. *Given the power series*

$$\sum_{k=0}^{\infty} a_k x^k,$$

one of the following occurs:

1. the series converges
2. $\exists R > 0$ such that the series converges absolutely for $|x| < R$ and diverges for $|x| > R$
3. The series converges absolutely for $x \in \mathbb{R}$.

Proof.

$$M = \{x \in \mathbb{R} : \text{series converges}\}.$$

$0 \in M$ by definition.

1. $M = 0$: Case 1.
2. M is unbounded: $\forall C > 0$ exists a $x \in M$ such that $|x| > C$. By Abel's Lemma, $M = \mathbb{R}$.
3. $M \neq 0$ but bounded: Let $R = \sup\{|x| : x \in M\}$. Abel's Lemma and definition of supremum we get case 2 of the theorem.

□

Definition 2.2 (Radius of Convergence). $R \in (0, \infty]$ is called the **Radius of Convergence** of

$$\sum_{k=0}^{\infty} a_k x^k$$

if for all $|x| \leq R$ the series converges.

Proposition 2.3. *Let*

$$\sum_{k=0}^{\infty} a_k x^k$$

be a power series with radius of convergence $R > 0$. Then the series converges uniformly on a compact subset of $|x| < R$.

Proof. Let $K \subset (-R, R)$ be compact. Pick $0 < r < R$ such that $K \subseteq [-r, r]$. We know that the series converges absolutely on $[-r, r]$. Since $|a_k x^k| \leq |a_k| r^k$. By the Weierstrass M test, the power series converges uniformly on K . □

Corollary 2.4. *Let*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

with radius of convergence of R . $f \in C((-R, R))$

Theorem 2.5 (Hadamard's Formula). *Let $(a_k)_{k=0}^{\infty}$ be the sequence of coefficients of a power series with radius R .*

$$R = \frac{1}{H}$$

where

$$H = \limsup_{k \rightarrow \infty} |a_k|^{\frac{1}{k}}$$

Proof. We have to show that H satisfies the properties of the radius of convergence: $|x| < \frac{1}{H} \Rightarrow$ the series converges absolutely, and if $|x| > \frac{1}{H}$ the series diverges. $|x| < \frac{1}{H} \Rightarrow |x|H \leq 1$. Thus $|x| \limsup |a_k|^{\frac{1}{k}} < 1 \Leftrightarrow L = \limsup |a_k x^k|^{\frac{1}{k}} < 1$. Thus $\forall \varepsilon > 0$ there exists a N such that $\forall k \geq N$, $|a_k x^k|^{\frac{1}{k}} < L + \varepsilon$. Take ε such that $L + \varepsilon < 1$. Then $|x_k x^k| < (L + \varepsilon)^k$. Thus for large enough k , we have a geometric series and thus the series converges absolutely.

Suppose $|x|H > 1$. Then $L > 1$. Then the terms $|a_k x^k|^{\frac{1}{k}} > 1$ for infinitely many k , thus $|a_k x^k| > 1$ for infinitely many k . Thus the series diverges. $\square$

Proposition 2.6. *Let $I \subset \mathbb{R}$ be an open interval. Let $f_n \in C^1(I)$ be such that $f_n \rightarrow f$ converges pointwise on I and $f_n' \rightarrow g$ locally uniformly (uniformly on compact subsets of I). Then $f \in C^1(I)$ and $f' = g$.*

Proof. Let $a \in I$. $\forall x \in I$:

$$\int_a^x f_n'(t) dt = f_n(x) - f_n(a).$$

Keep x fixed and $n \rightarrow \infty$:

$$f_n(x) - f_n(a) \rightarrow f(x) - f(a).$$

Since $f_n' \rightarrow g$ converge uniformly on compact subsets of I ,

$$\lim_{n \rightarrow \infty} \int_a^x f_n(t) dt = \int_a^x g(t) dt.$$

Note $g \in C(I)$, thus f is a $C^1(I)$. Thus $f'(x) = g$ by fundamental theorem of calculus. $\square$

Theorem 2.7. *Let*

$$\sum_{k=0}^{\infty} a_k x^k$$

be a power series with radius of convergence $R > 0$. Then

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

where $|x| < R$, satisfies: $f \in C^\infty((-R, R))$ and the derivatives can be obtained by differentiating the power series term by term.

Proof. The radius of convergence of the differentiated series:

$$\sum_{k=1}^{\infty} k a_k x^{k-1}$$

is equal to the radius of convergence of

$$\sum_{k=1}^n k a_k x^k$$

which is

$$\frac{1}{\limsup_{n \rightarrow \infty} |k a_k|^{\frac{1}{k}}} = \frac{1}{\limsup_{n \rightarrow \infty} |a_k|^{\frac{1}{k}}} = R.$$

□

3 Abel's Theorem

Theorem 3.1 (Abel's Continuity Theorem). *Let*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

with positive radius of convergence $0 < R < \infty$ and assume that

$$\sum_{k=0}^{\infty} a_k R^k$$

converges (not necessarily nicely). Then the convergence is uniform on $[0, R]$, then $f(x) \in C([0, R])$

Proof. For $x \in [0, R]$, let $t = \frac{x}{R} \Rightarrow x = Rt$. Thus the power series becomes a power series in t with radius of convergence of 1:

$$\sum_{k=0}^{\infty} a_k R^k t^k.$$

We also have convergence when $t = 1$. Suffices to treat the case where $R = 1$. We have

$$-\infty < \sigma = \sum_{k=0}^{\infty} a_k < \infty$$

$$\sigma_n = \sum_{k=1}^n a_k.$$

Consider for $x \in [0, 1)$,

$$\begin{aligned}
f(x) - \sum_{k=0}^n a_k x^k &= \sum_{k=n+1}^{\infty} a_k x^k \\
&= \sum_{k=n+1}^{\infty} (\sigma_k - \sigma_{k-1}) x^k \\
&= \sum_{k=n+1}^{\infty} ((\sigma_k - \sigma) - (\sigma_{k-1} - \sigma)) x^k \\
&= \sum_{k=n+1}^{\infty} (\sigma_k - \sigma) x^k - x \sum_{k=n+1}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} \\
&= \sum_{k=n+2}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} + (\sigma_n - \sigma) x^n - (\sigma_n - \sigma) x^n - x \sum_{k=n+1}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} \\
&= \sum_{k=n+1}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} - x \sum_{k=n+1}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} - (\sigma_n - \sigma) x^n \\
&= (1-x) \sum_{k=n+1}^{\infty} (\sigma_{k-1} - \sigma) x^{k-1} - (\sigma_n - \sigma) x^n
\end{aligned}$$

Let $\varepsilon > 0$. $\exists N$ such that

$$|\sigma_n - \sigma| \leq \varepsilon$$

for all $n \geq N$. We get for $n \geq N$ and $x \in [0, 1]$,

$$\begin{aligned}
\left| f(x) - \sum_{k=0}^n a_k x^k \right| &\leq \varepsilon x^n + (1-x) \sum_{k=n+1}^{\infty} \varepsilon x^{k-1} \\
&\leq \varepsilon + \varepsilon(1-x) \sum_{k=n}^{\infty} x^k \leq \varepsilon + \varepsilon x^n \leq 2\varepsilon
\end{aligned}$$

where $x \in [0, 1]$. Thus $\forall x \in [0, 1]$ this holds. Take

$$\sup_{x \in [0, 1]} \left| f(x) - \sum_{k=0}^n a_k x^k \right| \leq 2\varepsilon.$$

Thus the power series converges uniformly on $x \in [0, 1]$. □

Remark 3.1. The trick used here is called Abel's Partial summation.

Example 3.1. Consider $\log(1+x)$, it is given by the following Taylor series with radius of convergence of 1,

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k.$$

But when $x = 1$, the series converges (alternating). By Theorem 3.1, the RHS is continuous on $[0, 1]$. Letting $x \rightarrow 1^-$: we get

$$\log(2) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{x^k}$$

Example 3.2 (Basic Exam, Spring 2022). Let $(a_n)_{n=0}^{\infty}$ where $a_n \geq 0$ such that $na_n \rightarrow a \in (0, \infty)$. Look at

$$f(x) = \sum_{k=0}^{\infty} a_n x^k$$

with radius of convergence of 1.

$$\lim_{x \rightarrow 1^-} f'(x)(1-x) = a$$

Proof. We have

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}, |x| < 1$$

$$f'(x)(1-x)$$

□

4 Abel's Partial Summation

Consider: $\sum_{k=1}^n a_k b_k$ and let $B_k = \sum_{n=1}^{\infty} b_n$ and let $B_0 = 0$. Thus $b_k = B_k - B_{k-1}$.

Thus

$$\begin{aligned} \sum_{k=1}^n a_k b_k &= \sum_{k=1}^n a_k (B_k - B_{k-1}) \\ &= \sum_{k=1}^n a_k B_k - \sum_{k=0}^{n-1} a_{k+1} B_k \\ &= \sum_{k=1}^n a_k B_k - \sum_{k=1}^{n-1} a_{k+1} B_k \end{aligned}$$

We get:

$$\sum_{k=1}^n a_k b_k = a_n B_n + \sum_{k=1}^{n-1} (a_k - a_{k+1}) B_k$$

References